

Fellowship Centers Attendance Dashboard

Power BI Analytics Case Study

Executive Summary

A church operating over 1,000 fellowship centers across multiple regions required greater visibility into attendance trends, operational effectiveness, demographic participation, and evangelism outcomes. Although weekly attendance reports were being collected consistently, leadership lacked a consolidated mechanism for identifying growth opportunities, operational challenges, and regional performance differences.

To address this challenge, a Power BI solution was developed to transform almost two years of weekly attendance data into an interactive business intelligence platform.

The resulting dashboard enabled leadership to monitor key performance indicators, identify underperforming areas, evaluate outreach effectiveness, and support data-driven decision-making.

The Challenge

The church collected weekly attendance reports from hundreds of provinces across the city.

However, several challenges existed:

- Data was dispersed across multiple reports.
- Performance trends were difficult to identify.
- Leadership could not easily compare current performance with previous periods.
- Regional differences were not visible.
- Outreach effectiveness was difficult to measure objectively.

As a result, leadership decisions were largely based on individual reports rather than consolidated insights.

The objective of this project was to provide a single source of truth for monitoring growth and performance.

My Approach

I approached the project in four phases.

Phase 1 – Data Preparation

The raw attendance data was reviewed and transformed using Power Query.

Key activities included:

- Data validation
- Standardization of date fields



- Creation of Year and Month dimensions
- Development of a Calendar Table
- Removal of redundant fields

The final model supported accurate time-based analysis and trend reporting.

Phase 2 – KPI Design

The next step was identifying the metrics that matter most to leadership.

Rather than focusing solely on attendance totals, I developed KPIs that measure:

- Growth
- Operational effectiveness
- Outreach performance
- Participation patterns

Examples include:

- Year-over-Year Attendance Growth
- Center Activity Rate
- Attendance Rate per Meeting
- First Timer Ratio
- New Convert Ratio

Phase 3 – Dashboard Development

Five dashboards were created to address different leadership questions.

- Executive Summary
Provides a high-level view of organizational performance.
- Center Activity Analysis
Evaluates operational consistency across provinces and regions.
- Demographic Analysis
Examines participation patterns among males, females, and children.
- Attendance Analysis
Measures engagement and attendance effectiveness.
- Evangelism Analysis
Evaluates outreach effectiveness and visitor conversion performance.

Phase 4 – Data Storytelling

The final stage focused on translating metrics into actionable insights.

Rather than simply presenting charts, the dashboard was designed to answer strategic questions such as:

- Are we growing?
- Which regions require intervention?
- Are outreach efforts producing results?
- Where should leadership focus resources?



Key Insights

The analysis revealed several important patterns.

Insight 1: Center Activity Drives Attendance

Provinces generally recorded stronger attendance performance when fellowship centers are more active.

This suggests that operational discipline is a major contributor to growth.

Insight 2: Performance is uneven

Attendance growth and center activity varied significantly across regions.

A relatively small number of regions contributed a disproportionate share of overall performance.

This presents an opportunity to identify and replicate successful practices.

Insight 3: Assimilation Matters

Visitor growth alone does not guarantee sustainable growth.

The relationship between First Timers and New Converts highlighted the importance of follow-up and assimilation processes. Consistency in attendance of existing members is also important for growth to occur.

Insight 4: Demographics influence strategy

The demographic composition varied across regions.

Understanding these patterns can help leadership tailor engagement initiatives to local needs.

Business Recommendations

Based on the analysis, six strategic recommendations emerged.

- Improve center activity rates in underperforming provinces.
- Replicate successful practices from high-performing regions.
- Strengthen visitor assimilation and follow-up processes.
- Implement targeted interventions in low-performing areas.
- Develop demographic-focused engagement strategies.
- Embed dashboard metrics into regular leadership reviews.

Project Impact

This project transformed weekly attendance reporting into a decision-support system.

Leadership can now:

- Monitor growth trends in real time.
- Identify operational gaps quickly.
- Measure outreach effectiveness.
- Compare regional performance objectively.
- Make data-driven decisions with confidence.

The solution demonstrates how business intelligence can convert routine operational data into actionable organizational insights.



Skills Demonstrated

- Data Cleaning
- Power Query
- Data Modeling
- DAX
- Time Intelligence
- Exploratory Data Analysis
- KPI Design
- Data Visualization
- Business Intelligence
- Data Storytelling
- Executive Reporting
- Stakeholder Communication